

Notes: Irani, Iyer, Meisenzahl, and Peydro (RFS, 2021)

The Rise of Shadow Banking: Evidence from Capital Regulation

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1 Big picture

1.1 Question

Irani, Iyer, Meisenzahl, and Peydró study whether bank capital regulation helps explain the rise of shadow banks in the U.S. corporate loan market. The main question is:

When banks face binding or tightening capital constraints, do they sell syndicated corporate loan shares, and do lightly regulated nonbanks absorb those loans?

The paper then asks a second question:

If credit migrates from regulated banks to nonbanks, does that make credit markets more fragile in bad times?

The paper is therefore about both **regulatory arbitrage** and **financial stability**. Capital regulation may make banks safer individually, but it may also push credit risk into institutions with less stable funding.

1.2 Setting

The setting is the U.S. syndicated corporate loan market. This market is ideal for the paper because a single loan is commonly funded by several lenders. Some lenders are traditional banks, while others are nonbanks such as collateralized loan obligations, mutual funds, hedge funds, private equity funds, broker-dealers, insurance companies, pension funds, and finance companies.

The key data source is the **Shared National Credit Program** (SNC), a supervisory credit register. The SNC is unusually useful because it observes the ownership of syndicated loan shares over time. This means the authors can see not only who funds a loan at origination, but also whether a bank later sells part or all of its loan share in the secondary market.

1.3 Core mechanism

The mechanism is simple.

1. A bank with low regulatory capital wants to raise its capital ratio.
2. One way to do this is to reduce risk-weighted assets.
3. Selling a corporate loan share reduces loan exposure and therefore risk-weighted assets.
4. Nonbanks, which are less tightly regulated and do not face the same capital requirements, can buy the loan share.
5. The credit risk moves from the regulated banking sector into the shadow banking sector.

Interpretation. The paper is not saying that all nonbank lending is bad. The point is that bank regulation can change *who holds credit risk*. The financial stability consequences depend on whether the new holders have stable funding during stress.

1.4 Main empirical result

The paper finds strong evidence that banks with weaker capital positions sell more syndicated term loan shares. This pattern is especially strong when capital is scarce and when loans have higher capital requirements.

Key results:

- Lower Tier 1 capital is associated with a higher probability that a bank sells a loan share.
- The effect is stronger when the TED spread is high, which proxies for tighter funding conditions.
- The effect is much stronger for nonperforming loans, which carry higher expected losses and higher capital charges.
- At the loan level, lower average bank capital in the syndicate predicts a higher nonbank funding share.
- A one-standard-deviation decrease in bank capital raises the nonbank share by about **3.25 percentage points**, equal to about **14.1% of the mean** nonbank share.

1.5 Identification in one sentence

The main identification idea is to compare banks **within the same loan-year**. If two banks hold shares of the same loan at the same time, then borrower quality, loan quality, and loan-time shocks are held fixed. Differences in selling behavior can then be related to differences in bank capital.

1.6 Capital regulation shock

The authors also use plausibly exogenous variation from the U.S. implementation of Basel III. The international Basel III agreement was known, but the precise U.S. implementation contained surprise features, including especially punitive treatment of mortgage servicing rights and changes in risk-weighting rules. These features created unexpected capital shortfalls for some banks.

The shock-based evidence confirms the baseline result: banks with larger Basel III capital shortfalls sell more loan shares, and nonbanks increase their holdings in affected loan syndicates.

1.7 Financial stability result

The second part of the paper studies the 2008 crisis. Loans with higher precrisis nonbank funding experienced weaker credit availability and larger price declines during the crisis. The negative effects are concentrated among loans funded by **unstable nonbanks**: broker-dealers, hedge funds, private equity funds, mutual funds, and related investment funds. These institutions are more exposed to redemption risk, short-term funding pressure, or market liquidity pressure.

Main crisis results:

- Higher precrisis nonbank share predicts lower loan-level and firm-level credit growth during 2007–2008.
- Higher precrisis nonbank share predicts a greater probability that the loan exits the SNC sample

by the end of 2008.

- A ten percentage point increase in unstable nonbank funding raises the loan exit rate by about **4.18 percentage points**.
- A one-standard-deviation higher nonbank share accounts for about **19.2%** of the mean loan price fall during the crisis.
- Stable nonbanks, such as insurance companies and pension funds, do not generate the same negative effects.

1.8 Broader takeaway

The paper's main takeaway is a regulatory tradeoff. Bank capital regulation can make regulated banks safer, but it can also push risky credit outside the regulated banking system. If the new lenders have fragile funding, then the system may become more vulnerable during crises.

2 Data and measurement

2.1 Main data: Shared National Credit Program

The main data source is the Shared National Credit Program. The SNC is a supervisory credit register maintained by U.S. bank regulators. It covers large syndicated credits, including loan commitments larger than \$20 million that are shared by multiple institutions.

The SNC records:

- borrower identity;
- loan origination date and maturity date;
- loan type, especially term loans and credit lines;
- the administrative agent or lead arranger;
- all participating lenders;
- each lender's loan share or exposure;
- regulatory loan quality classifications;
- whether each lender is a bank or nonbank.

The authors use the data after 1993, when data quality is highest, and follow loans through 2014.

2.2 Unit of observation

The central unit of observation is a **loan share–lender–year** triple. A syndicated loan may have many lenders, and each lender owns a loan share. Because the SNC tracks the same loan over time, the authors can observe whether a lender increases, reduces, exits, or newly enters a loan syndicate.

For the main bank-capital tests, the sample is restricted to loan shares funded by U.S. banks and includes:

- **20,685** unique syndicated loans;
- **161,794** loan share–lender–year observations;
- **1,897** banks;
- sample period **1993–2014**.

2.3 Loan sales

The paper defines a loan sale as follows:

A lender sells a loan share in year t if its exposure to that loan is lower in year t than in year $t - 1$.

The sale can be a complete exit or a partial reduction in exposure. The authors exclude maturing loans so that normal repayment is not misclassified as a sale. They also code sales at the bank holding company level, so transfers within the same bank organization are not counted as true sales.

$$\text{LoanSale}_{ijt} = 1\{\text{bank } j \text{ reduces its stake in loan } i \text{ during year } t\}.$$

Interpretation. This variable captures secondary-market risk transfer. It is central because the paper is about whether capital-constrained banks remove corporate credit risk from their balance sheets.

2.4 Nonbank share

The loan-level nonbank share is the fraction of a loan’s outstanding funding held by nonbanks:

$$\text{NonbankShare}_{it} = \frac{\text{loan amount held by nonbanks in loan } i \text{ at time } t}{\text{total loan amount outstanding in loan } i \text{ at time } t}.$$

Interpretation. This is the paper’s main measure of shadow-bank entry. A higher nonbank share means that more of the loan is funded outside the traditional banking sector.

2.5 Nonbank classification

The SNC classifies lenders broadly as domestic banks, foreign banks, domestic nonbanks, and foreign nonbanks. The authors further classify nonbanks into more detailed groups:

- collateralized loan obligations (CLOs);
- hedge funds and private equity funds;

- mutual funds and other investment funds;
- broker-dealers;
- insurance companies;
- pension funds;
- finance companies;
- other domestic, foreign, or unclassified nonbanks.

For crisis tests, nonbanks are divided by liability fragility:

- **Stable nonbanks:** insurance companies and pension funds.
- **Unstable nonbanks:** broker-dealers, hedge funds, private equity funds, mutual funds, and related investment funds.

CLOs are not always assigned to stable or unstable categories because their liability maturity and funding structure are not observed in enough detail.

2.6 Bank capital and bank controls

The authors merge the SNC data with bank balance-sheet data from regulatory reports. The main bank capital measure is:

$$\text{Tier1Capital/RWA}_{j,t} = \frac{\text{Tier 1 capital}_{j,t}}{\text{risk-weighted assets}_{j,t}}.$$

They also use alternative capital measures:

- Tier 1 capital gap;
- total capital to risk-weighted assets;
- Tier 1 leverage ratio;
- Basel III capital shortfall in the regulatory-shock analysis.

Bank controls include bank size, wholesale funding dependence, real estate loan share, C&I loan share, noninterest income, the loan share relative to bank assets, and whether the bank is the lead arranger.

2.7 Secondary market price data

For price effects during the crisis, the authors use Loan Syndication and Trading Association (LSTA) Mark-to-Market data. These data provide bid and ask quotes for traded syndicated loans. The authors match a subset of SNC loans to LSTA prices and measure the loan price as the midpoint of the bid and ask quotes.

$$\text{LoanPrice}_{it} = \frac{\text{Bid}_{it} + \text{Ask}_{it}}{2}.$$

The main price outcome is the annual change in loan price from 2007 to 2008.

2.8 Summary facts

Several descriptive facts motivate the paper.

- Nonbank funding of syndicated term loans rose from about **20% in 1993** to about **70% in 2014**.
- Nonbank entry accelerated between **2002 and 2006**.
- CLOs became the largest nonbank investor class in the early 2000s.
- After 2008, hedge funds, private equity funds, and loan mutual funds became increasingly important.
- In the loan share–bank–year sample, a loan share exposure is reduced in about **37%** of observations.
- The average loan-level nonbank share is about **23.1%**.

3 Models and empirical specifications

3.1 Baseline loan-sale specification

The baseline specification is a linear probability model:

$$\text{LoanSale}_{ijt} = \alpha_{it} + \alpha_j + \beta \text{Tier1Capital/RWA}_{j,t-1} + \gamma X_{ij,t-1} + \varepsilon_{ijt}. \quad (1)$$

Here:

- i indexes loans;
- j indexes banks;
- t indexes years;
- LoanSale_{ijt} equals one if bank j reduces its exposure to loan i in year t ;
- α_{it} are loan-year fixed effects;
- α_j are bank fixed effects;
- $\text{Tier1Capital/RWA}_{j,t-1}$ is the lagged Tier 1 capital ratio;
- $X_{ij,t-1}$ contains bank and loan-share controls.

Interpretation. The coefficient β asks whether banks with lower capital sell more loan shares, comparing banks that hold the same loan in the same year. If capital-constrained banks sell loans to reduce risk-weighted assets, then $\beta < 0$.

3.2 Why loan-year fixed effects matter

Loan-year fixed effects absorb all factors common to a particular loan in a particular year. This includes borrower quality, loan quality, loan demand, loan maturity, industry shocks, and any time-varying loan-level default risk.

Interpretation. The identification comes from comparing two banks that are exposed to the same loan at the same time. If one bank has lower capital and sells while the other does not, the paper attributes the difference to bank-side balance-sheet incentives rather than borrower-side fundamentals.

3.3 Dynamic capital scarcity specification

To test whether capital matters more when funding markets are tight, the authors interact bank capital with the TED spread:

$$\text{LoanSale}_{ijt} = \alpha_{it} + \alpha_j + \beta_1 \text{Tier1Capital/RWA}_{j,t-1} + \beta_2 \left(\text{Tier1Capital/RWA}_{j,t-1} \times \text{TED}_t \right) + \gamma X_{ij,t-1} + \varepsilon_{ijt}. \quad (2)$$

Interpretation. The TED spread proxies for marketwide funding stress. A negative β_2 means that the effect of low bank capital on loan sales becomes stronger when external funding is harder to obtain.

3.4 Regulatory rating split

The authors estimate the baseline model separately for loans rated **pass** and loans rated **fail**. A fail rating includes special mention, substandard, doubtful, or loss.

Interpretation. Nonperforming loans have higher expected losses and capital charges. If regulatory capital is the mechanism, low-capital banks should be especially eager to sell failed loans. This is exactly what the paper finds.

3.5 Nonbank-entry specification

The paper then moves from the bank-loan-share level to the loan-year level. The dependent variable becomes the share of the loan held by nonbanks:

$$\text{NonbankShare}_{it} = \alpha_t + \beta \overline{\text{Tier1Capital/RWA}}_{i,t-1} + \gamma \overline{X}_{i,t-1} + \delta Z_{it} + u_{it}. \quad (3)$$

Here $\overline{\text{Tier1Capital/RWA}}_{i,t-1}$ is an aggregate measure of bank capital among the banks in the syndicate. The baseline aggregation is the equally weighted average across bank syndicate members.

The authors also use value-weighted averages, medians, dominant-bank averages, and lead-arranger capital.

Interpretation. If low-capital banks sell and nonbanks buy, then loans whose bank syndicates have lower capital should have higher nonbank shares. The coefficient should therefore be negative.

3.6 Basel III capital-shortfall specification

For the regulatory shock analysis, the authors use the 2012 U.S. Basel III implementation. A simplified version of the loan-sale regression is:

$$\text{LoanSale}_{ij,2012Q3} = \alpha_i + \beta \text{BaselIIIShortfall}_{j,2012Q2} + \theta \text{Tier1Capital/RWA}_{j,2012Q2} + \gamma X_j + \varepsilon_{ij}. \quad (4)$$

Here α_i is a loan fixed effect, and $\text{BaselIIIShortfall}_{j,2012Q2}$ is the bank-level difference between current Tier 1 capital under Basel I and the proposed Tier 1 capital level under Basel III.

Interpretation. The shortfall is more negative for banks hit harder by the U.S. Basel III rule. A negative coefficient means that banks with more severe capital shortfalls are more likely to sell loan shares.

At the loan-syndicate level, the authors also estimate:

$$\Delta \text{NonbankShare}_i = \alpha + \beta \max_j \{ \text{BaselIIIShortfall}_j \} + \gamma X_i + u_i. \quad (5)$$

Interpretation. This asks whether loans exposed to banks with larger Basel III capital shortfalls experience larger increases in nonbank funding after the regulatory shock.

3.7 Credit availability during the crisis

The main crisis credit-growth measure is symmetric credit growth:

$$\text{CreditGrowth}_i = \frac{\text{Credit}_{i,2008} - \text{Credit}_{i,2006}}{0.5\text{Credit}_{i,2008} + 0.5\text{Credit}_{i,2006}}. \quad (6)$$

The authors also define an exit indicator:

$$\text{Exit}_i = 1\{\text{loan } i \text{ is no longer in the SNC sample by 2008:Q4}\}.$$

The cross-sectional crisis specification is:

$$\Delta \text{Credit}_i = \alpha + \beta \text{NonbankShare}_{i,2006:Q4} + \gamma X_{i,2006:Q4} + \varepsilon_i. \quad (7)$$

Interpretation. The coefficient β asks whether loans with more nonbank funding before the crisis experienced sharper contractions in credit availability during the crisis.

3.8 Stable versus unstable nonbank funding

The authors decompose nonbank funding into stable and unstable shares:

$$\Delta\text{Credit}_i = \alpha + \beta_U \text{UnstableNB}_{i,2006:Q4} + \beta_S \text{StableNB}_{i,2006:Q4} + \gamma X_i + \varepsilon_i. \quad (8)$$

Interpretation. The theory predicts that fragile funding matters most. Therefore β_U should be negative for credit growth and positive for exit, while β_S should be small or insignificant.

3.9 Loan price decline specification

For traded loans, the authors estimate:

$$\Delta\text{LoanPrice}_i = \alpha + \beta \text{NonbankShare}_{i,2006:Q4} + \gamma X_{i,2006:Q4} + \varepsilon_i. \quad (9)$$

They also use the stable/unstable decomposition.

Interpretation. A negative β means that loans with more precrisis nonbank funding experienced larger secondary-market price declines during the 2008 crisis. If unstable nonbanks drive this effect, that supports a fire-sale or funding-fragility interpretation.

4 Identification and instruments

4.1 Main identification strategy

The strongest part of the paper’s baseline design is the loan-year fixed effect. Syndicated loans naturally have multiple lenders. Therefore, the authors can compare selling decisions across banks inside the same loan syndicate in the same year.

This design removes:

- borrower-level credit demand;
- borrower quality;
- loan quality;
- industry shocks to the borrower;
- time-varying risk of the specific loan;
- refinancing or maturity pressure common to the loan-year.

Interpretation. The model is a within-loan comparison. It asks: among banks holding the same loan at the same time, does the lower-capital bank sell more often?

4.2 Key identifying assumptions

The paper relies on two main assumptions for the baseline result.

Assumption 1: loan sales are supply-side decisions. Borrowers cannot selectively force low-capital banks to leave or high-capital banks to stay. This is plausible because syndicated term loan shares can trade in secondary markets, and borrowers have limited influence over the identity of secondary-market buyers and sellers after origination.

Assumption 2: bank capital is not only proxying for omitted bank-level shocks. For example, if low-capital banks also have bad risk-management systems or expect losses in unrelated portfolios, those factors might explain loan sales. The paper uses bank fixed effects, bank controls, robustness tests, and the Basel III shock to address this concern.

4.3 Robustness tests for the loan-sale result

The baseline result survives many checks:

- excluding finance, insurance, real estate, and construction borrowers;
- restricting to loan-years with no amendments or refinancing;
- using credit lines as a placebo, where the secondary market is thinner;
- using alternative timing and sample windows;
- dropping fixed effects to test sensitivity to omitted variables;
- using alternative measures of loan sales;
- using alternative capital measures such as Tier 1 gap and total capital/RWA;
- checking that credit default swaps do not explain the behavior;
- showing that the effect is stronger for weak bank-borrower relationships.

Interpretation. These tests reinforce the regulatory-capital interpretation. Low-capital banks sell assets that are easier to offload and less relationship-sensitive.

4.4 Why credit lines are a useful placebo

Credit lines are generally retained by banks because banks have a comparative advantage in managing liquidity risk. The secondary market for credit lines is much less active than the secondary market for term loans.

Interpretation. If the paper’s result were simply “low-capital banks sell everything,” we might expect a similar result for credit lines. The fact that the credit-line result is weak supports the idea that term-loan secondary-market liquidity is essential to the mechanism.

4.5 Basel III as a regulatory shock

The U.S. implementation of Basel III provides a more direct regulatory-capital shock. The global Basel III framework was known, but the U.S. version had important and partly unexpected features. In particular:

- it changed what counted as Tier 1 capital;

- it treated mortgage servicing rights more harshly than many banks expected;
- it adjusted risk weights for certain exposures;
- it created heterogeneous capital shortfalls across banks.

The important feature is that banks with similar pre-rule capital ratios could face different Basel III shortfalls because of their exposures to the rule's specific implementation details.

4.6 Basel III identification logic

The Basel III test compares banks holding the same loan around the regulatory announcement. If one bank receives a larger unexpected capital shortfall and sells more, this is difficult to explain using borrower demand or loan quality, because those are common across banks inside the same loan.

Interpretation. The Basel III analysis is not a pure randomized experiment, but it sharply improves identification because the capital shock is tied to regulatory implementation quirks rather than directly to syndicated-loan fundamentals.

4.7 Crisis-effect identification

The crisis tests are more cross-sectional and therefore less internally clean than the within-loan bank-sale tests. The authors ask whether loans with more nonbank funding before the crisis perform worse during the crisis.

The concern is selection: perhaps nonbanks chose riskier loans before the crisis, and those loans would have performed badly anyway.

The authors address this concern in several ways:

- they control for loan size, syndicate size, borrower industry, remaining maturity, and regulatory downgrades;
- they control for average bank characteristics in the syndicate;
- they use firm-level tests to see whether borrowers substitute to other syndicated loans;
- they compare stable and unstable nonbanks;
- they examine observable borrower financial condition and ex post performance in auxiliary tests.

Interpretation. The crisis tests should be read as evidence consistent with a funding-fragility channel, not as a perfect randomized causal estimate. The stable-versus-unstable split is especially important because it links the adverse outcomes to liability fragility rather than to nonbank status alone.

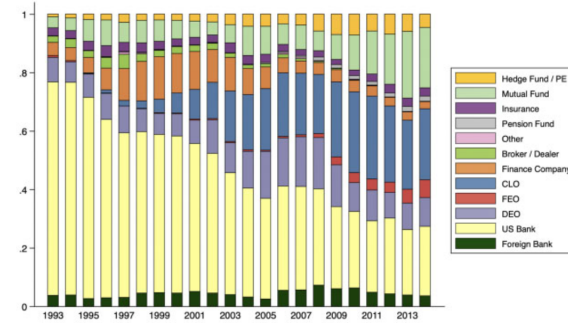


Figure 1
U.S. syndicated term loan funding market share by entity type (1993–2014)
The categories in the figure refer to groups of financial firms and, to ensure confidentiality, data for no individual firm are disclosed. “DEO,” “FEO,” and “Other” denote nonbank entities with a domestic, foreign, and unknown origin, respectively. These nonbank lenders could not be classified (into any of the other categories) based on our lender lists.

5 Tables and figures

5.1 Figure 1: Nonbank funding in syndicated term loans

Read: Figure 1 plots the composition of syndicated term loan funding by entity type. Bank funding falls over time, while nonbank funding rises from about 20% in 1993 to about 70% in 2014. CLOs become especially important in the early 2000s. After the crisis, hedge funds, private equity, and mutual funds become more prominent.

Economic message: This figure motivates the paper. The rise of nonbanks is not a marginal phenomenon. By the end of the sample, nonbanks fund most syndicated term-loan credit. The empirical question is whether bank capital regulation helps explain this shift.

5.2 Figures 2 and 3: Secondary-market sales and purchases

Read: Figures 2 and 3 show secondary-market sales and purchases of syndicated term loan shares. Nonbanks begin to dominate secondary-market trading around the early 2000s, and their trading activity becomes especially large after 2007.

Economic message: The key point is that nonbank entry is not only about primary issuance. A large part of the migration from banks to nonbanks occurs through secondary-market trading. This is why the SNC ownership panel is essential.

5.3 Table 2: Summary statistics

Read: Table 2 reports summary statistics for the main loan-share sample. Loan shares are reduced in 37% of observations, and complete exits occur in 6.5% of observations. The average loan commitment is about \$275 million. The average loan-level nonbank share is 23.1%. Below-median-capital banks have lower Tier 1 capital ratios, larger balance sheets, higher wholesale funding shares, and larger C&I loan shares.

Economic message: The table shows that loan-share trading is frequent and that banks differ meaningfully in regulatory capital. This creates enough variation to study whether bank balance-

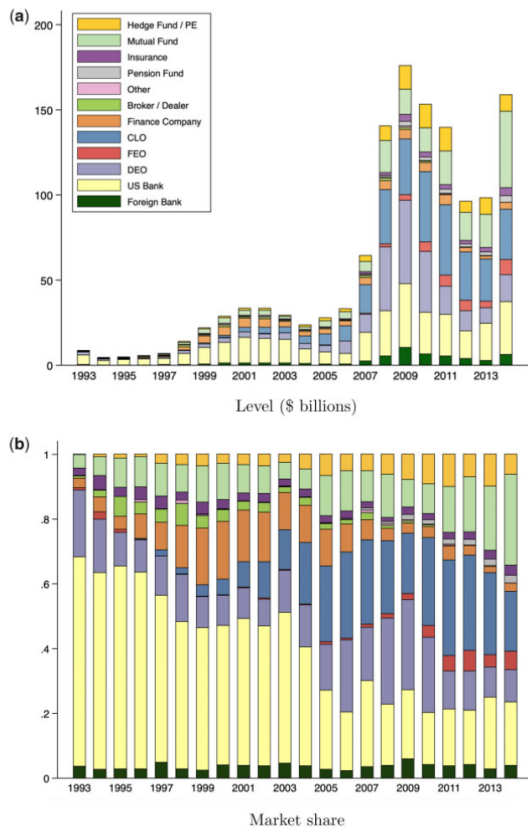


Figure 2
Secondary market sells of U.S. syndicated term loan shares (1993–2014)
 Loan share sales in levels (\$ billions, top panel) and by market share (bottom panel). A loan share is a fraction of a syndicated loan commitment. A loan share sale occurs when a lender reduces its ownership stake in a loan share relative to the previous year. The categories in the figure refer to groups of financial firms and, to ensure confidentiality, data for no individual firm are disclosed. “DEO,” “FEO,” and “Other” denote nonbank entities with a domestic, foreign, and unknown origin, respectively. These nonbank lenders could not be classified (into any of the other categories) based on our lender lists.

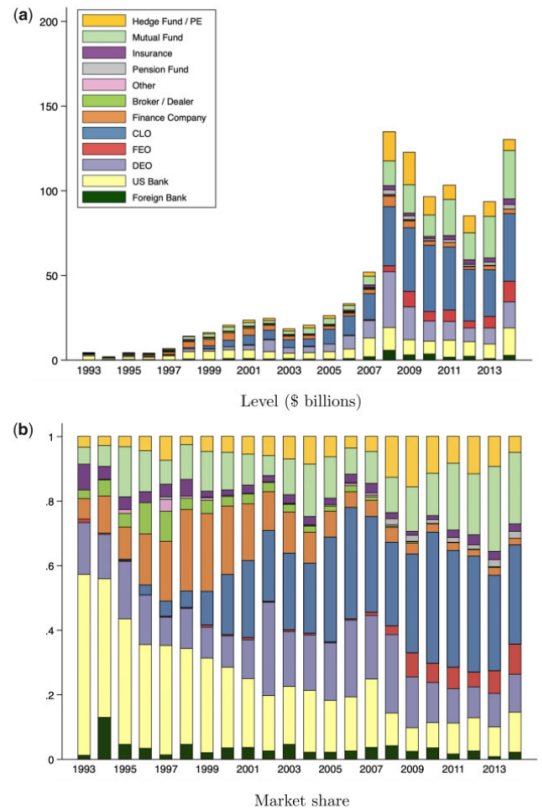


Figure 3
Secondary market buys of U.S. syndicated term loan shares (1993–2014)
 Loan share buys in levels (\$ billions, top panel) and by market share (bottom panel). A loan share is a fraction of a syndicated loan commitment. A loan share buy occurs when a lender increases its ownership stake in a loan share relative to the previous year. The categories in the figure refer to groups of financial firms and, to ensure confidentiality, data for no individual firm are disclosed. “DEO,” “FEO,” and “Other” denote nonbank entities with a domestic, foreign, and unknown origin, respectively. These nonbank lenders could not be classified (into any of the other categories) based on our lender lists.

sheet constraints affect loan sales and nonbank entry.

Table 2
Summary statistics for banks and loan sales tests

	N	Mean	Std.	p25	Med.	p75	N	Mean	Std.	p25	Med.	p75
	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]	[10]	[11]	[12]
Loan-level variables												
<i>Loan Sale</i>	161,794	0.370	0.483	0	0	1						
<i>Loan Share/Assets</i>	161,794	0.007	0.002	0.000	0.000	0.000						
<i>Loan Size</i>	161,794	274.0	619.9	34.5	95.0	256.0						
<i>Lead Arranger</i>	161,794	0.181	0.385	0	0	0						
<i>Non-Bank Share</i>	39,058	0.231	0.320	0	0	0.403						
Bank-level variables												
		Below-median capital						Above-median capital				
<i>Tier 1 Capital/RWA</i>	2,017	0.100	0.014	0.002	0.101	0.112	2,018	0.175	0.060	0.125	0.133	0.191
<i>Tier 1 Gap</i>	2,017	-0.009	0.020	-0.022	-0.011	0.003	2,018	0.006	0.040	-0.018	0.000	0.023
<i>Total Capital/RWA</i>	2,017	0.115	0.012	0.107	0.115	0.124	2,018	0.187	0.061	0.147	0.166	0.203
<i>Tier 1 Leverage</i>	2,017	0.078	0.014	0.060	0.078	0.086	2,018	0.109	0.035	0.087	0.100	0.119
<i>Bank Size</i>	2,017	20.91	1.964	19.52	20.81	22.12	2,018	19.48	1.747	18.45	19.45	20.76
<i>Wholesale Funding</i>	2,017	0.300	0.146	0.192	0.283	0.389	2,018	0.231	0.147	0.126	0.202	0.297
<i>Real Estate Loan Share</i>	2,017	0.607	0.104	0.496	0.637	0.753	2,018	0.631	0.217	0.513	0.685	0.795
<i>C&I Loan Share</i>	2,017	0.116	0.101	0.011	0.110	0.170	2,018	0.062	0.086	0	0.015	0.101
<i>Non-Interest Income</i>	2,017	0.154	0.099	0.088	0.136	0.195	2,018	0.153	0.123	0.075	0.121	0.192

The sample is restricted to loans held by at least two U.S. banks with valid covariates at the beginning of the year. Loan-level variables are averaged (unweighted) across loan share-years. Bank-level variables are averaged across bank-years. Bank-level summary statistics split by above- and below-median beginning-of-year *Tier 1 Capital/RWA*. The sample period is from 1993 to 2014. All variables are defined in Table A1.

Table 3
Bank regulatory capital and syndicated loan sales

Dependent variable: $Loan\ Sales_{ijt}$	Baseline [1]	Dynamic [2]	Regulatory rating	
			Pass [3]	Fail [4]
<i>Tier 1 Capital/RWA_{t-1}</i>	-0.158*** (0.057)	-0.189*** (0.050)	-0.108* (0.060)	-0.499** (0.196)
<i>Tier 1 Capital/RWA_{t-1} × TED_t</i>		-0.292*** (0.070)		
<i>Size_{t-1}</i>	-0.004 (0.004)	0.005 (0.003)	-0.002 (0.004)	-0.012 (0.012)
<i>Wholesale Funding_{t-1}</i>	0.110*** (0.017)	0.100*** (0.014)	0.111*** (0.018)	0.121** (0.057)
<i>Real Estate Loan Share_{t-1}</i>	0.020 (0.019)	0.043*** (0.017)	0.027 (0.020)	-0.036 (0.062)
<i>C&I Loan Share_{t-1}</i>	-0.119*** (0.030)	-0.052** (0.026)	-0.076** (0.031)	-0.303*** (0.004)
<i>Non-Interest Income_{t-1}</i>	0.009 (0.018)	-0.003*** (0.000)	-0.001*** (0.001)	-0.003*** (0.001)
<i>Loan Share/Assets_{t-1}</i>	0.006*** (0.001)	0.005*** (0.001)	0.006*** (0.002)	0.008 (0.005)
<i>Lead Arranger_{t-1}</i>	-0.028*** (0.003)	-0.027*** (0.003)	-0.026*** (0.003)	-0.033*** (0.009)
Bank controls × <i>TED_t</i>	N	Y	N	N
Bank fixed effects	Y	Y	Y	Y
Loan-year fixed effects	Y	Y	Y	Y
Observations	97,238	97,238	83,759	13,479
R ²	0.878	0.873	0.881	0.870

This table shows the effects of bank regulatory capital for loan sales. The unit of observation in each regression is a loan share-bank-year triple. The dependent variable is an indicator variable equal to one if a lender reduces its ownership stake in a loan that it funded in the previous year. Column [1] includes the sample of loan sales from 2002 to 2014. Column [2] interacts capital with the TED spread (TED_t), defined as the yearly average of the daily difference between the three-month London Interbank Offered Rate (LIBOR) and the three-month U.S. Treasury rate. Note that TED_t is demeaned. Columns [3] and [4] classify a loan as "Pass" by the examining agency if it has not been criticized in any way and "Fail" otherwise (i.e., the loan is rated special mention, substandard, doubtful, or loss). All columns include controls for bank and loan-year fixed effects, and an indicator variable for whether the bank has undergone a merger in the past year. All variables are defined in Table A1. Standard errors (in parentheses) are clustered at the loan level. ***, **, and * denote 1%, 5%, and 10% statistical significance, respectively.

5.4 Table 3: Bank capital and loan sales

Read: Table 3 estimates the baseline loan-sale equation. The coefficient on Tier 1 Capital/RWA is negative and statistically significant. In the baseline column, the coefficient is about -0.158 . The effect becomes stronger when interacted with the TED spread. The effect is also stronger for failed loans than for pass loans: the coefficient is around -0.499 for failed loans versus about -0.108 for pass loans.

Economic message: Low-capital banks sell more loan shares. The fact that the effect is stronger during funding stress and for nonperforming loans strongly supports the interpretation that banks sell loans to reduce risk-weighted assets and protect capital ratios.

5.5 Table 4: Further loan-sale tests

Read: Table 4 checks the robustness of the loan-sale result. The result survives excluding FIRE borrowers, focusing on loan-years with no contract amendment, using alternative loan-sale measures, and using alternative capital measures. The placebo test on credit lines does not produce the same negative capital effect.

Economic message: The robustness checks show that the result is not simply driven by borrower distress, loan renegotiation, real estate exposure, or general bank weakness. The term-loan secondary market is the channel through which capital-constrained banks can offload risk.

5.6 Table 5: Nonbank entry

Read: Table 5 collapses the data to the loan-year level and uses nonbank share as the dependent variable. The capital variable is the average Tier 1 capital ratio of banks in the syndicate. The

Table 4
Bank capital and loan sales: Further tests
Panel A: Specification checks

Dependent variable: $Loan\ Sales_{ijt}$	Exclude FIRE [1]	No Amend [2]	Credit lines [3]	Alternate timing [4]	Exclude fixed effects [5]
$Tier\ 1\ Capital/RWA_{t-1}$	-0.179*** (0.061)	-0.151** (0.060)	0.051 (0.037)	-0.044 (0.027)	-0.198*** (0.054)
Bank fixed effects	Y	Y	Y	Y	N
Loan-year fixed effects	Y	Y	Y	Y	N
Observations	83,707	87,510	343,241	161,794	97,238
R^2	0.878	0.878	0.712	0.860	0.100

Panel B: Alternative measurement of loan sales

Dependent variable:	$Loan\ Share_{ijt}/Assets_{i,t-1}$		$Loan\ Sale\ Amount_{ijt}/Assets_{i,t-1}$		
Size-based classification:	None [1]	None [2]	Below med. [3]	Above med. [4]	Top dec. [5]
$Tier\ 1\ Capital/RWA_{t-1}$	4.030*** (0.347)	2.153*** (0.281)	-0.094** (0.045)	-0.095* (0.053)	0.035 (0.054)
Bank fixed effects	N	Y	Y	Y	Y
Loan-year fixed effects	Y	Y	Y	Y	Y
Observations	161,794	161,794	74,321	74,213	60,320
R^2	0.635	0.860	0.882	0.850	0.768

(Continued)

Table 4
(Continued)

Panel C: Alternative measurement of regulatory capital

Dependent variable: $Loan\ Sales_{ijt}$	$Tier\ 1\ Gap_{t-1}$				$Total\ Capital/RWA_{t-1}$			
Regulatory capital measure:	Regulatory rating				Regulatory rating			
	Baseline [1]	Dynamic [2]	Pass [3]	Fail [4]	Baseline [5]	Dynamic [6]	Pass [7]	Fail [8]
$Capital_{t-1}$	-0.469*** (0.077)	-0.314*** (0.079)	-0.479*** (0.082)	-0.470* (0.256)	-0.171*** (0.047)	-0.185*** (0.047)	-0.127*** (0.049)	-0.484*** (0.148)
$Capital_{t-1} \times TED_t$		-0.698*** (0.118)				-0.306*** (0.073)		
Bank controls	Y	Y	Y	Y	Y	Y	Y	Y
Bank controls $\times\ TED_t$	N	Y	N	N	N	Y	N	N
Bank fixed effects	Y	Y	Y	Y	Y	Y	Y	Y
Loan-year fixed effects	Y	Y	Y	Y	Y	Y	Y	Y
Observations	97,238	97,238	83,759	13,479	97,238	97,238	83,759	13,479
R^2	0.872	0.873	0.876	0.854	0.872	0.873	0.878	0.854

This table shows robustness checks for the effects of bank regulatory capital for loan sales. The unit of observation in each regression is a loan share-bank-year triple. The dependent variable is an indicator variable equal to one if a lender reduces its ownership stake in a loan that is funded in the previous year. In panel A, Column [1] excludes loans made to finance, insurance, and real estate sectors. Column [2] restricts the sample to loans years in which no contract amendment or refinancing took place during the year. Column [3] includes credit line loan shares in the sample. Column [4] examines the extended time period, including from 1993 to 2001, where the loan secondary market was less active. Column [5] drops the bank and loan-year fixed effects. Panel B examines alternative measures of loan sales. In columns [1] and [2], the dependent variable is the loan size in dollars scaled by bank assets at the end of the previous year. In columns [3] to [5], the numerator is instead the dollar value of the loan share sold scaled by bank assets. Here, we separately consider sales that are small (below median loan sale size), large (above median), and the largest (top decile). Panel C examines alternative measures of bank regulatory capital as independent variables and repeats the tests described in Table 3. All columns include the bank controls shown in Table 3, controls for bank and loan-year fixed effects, and an indicator variable for whether the bank has undergone a merger in the past year. All variables are defined in Table A1. Standard errors (in parentheses) are clustered at the loan level. ***, **, and * denote 1%, 5%, and 10% statistical significance, respectively.

coefficient on bank capital is negative across specifications. A one-standard-deviation decline in bank capital corresponds to a 3.25 percentage point increase in nonbank share, equal to 14.1% of the mean.

Economic message: This table closes the loop between bank sales and shadow-bank entry. Capital-constrained banks do not only sell to other banks. Their loan syndicates end up with more nonbank funding.

Table 5

Nonbank entry

Dependent variable: $Nonbank\ Share_{it}$

Syndicate aggregation: Mean (EW)

Regulatory rating:

	All [1]	All [2]	All [3]	Fail [4]	Fail [5]
$Tier\ 1\ Capital/RWA_{t-1}$	-1.547*** (0.470)	-1.582** (0.640)	-1.460*** (0.183)	-1.406*** (0.304)	-1.025*** (0.316)
$Tier\ 1\ Capital/RWA_{t-1} \times TED_t$			-2.954*** (0.601)		-4.655*** (0.980)
Bank controls	Y	Y	Y	Y	Y
Loan controls	N	Y	Y	Y	Y
Year fixed effects	Y	Y	Y	Y	Y
Observations	39,058	29,121	29,121	5,380	5,380
R^2	0.102	0.203	0.210	0.266	0.270

This table shows the effects of bank regulatory capital for loan acquisition by nonbanks. The unit of observation in each regression is a loan-year. The dependent variable is the fraction of the loan held by nonbanks. Columns [3] and [5] interact bank capital with the TED spread (TED_t), which is defined as the yearly average of the daily difference between the three-month London Interbank Offered Rate (LIBOR) and the three-month U.S. Treasury rate. Note that TED_t is demeaned. Columns [4] and [5] consider loans that have been classified as "Fail" by the examining agency. These are loans rated special mention, substandard, doubtful, or loss. Where indicated, independent variables—bank controls shown in Table 3—are coded at the loan syndicate level by taking the simple (equally weighted) average across syndicate member banks. The sample period is from 1993 to 2014. Where indicated, the columns include controls for bank, loan, and year fixed effects, and loan controls (a regulatory pass/fail dummy and the natural logarithm of loan maturity). All variables are defined in Table A1. Standard errors (in parentheses) are clustered at the year level. ***, **, and * denote 1%, 5%, and 10% statistical significance, respectively.

Table 6

Nonbank entry: Further tests

Dependent variable: $Nonbank\ Share_{it}$

Robustness test:

	Syndicate aggregation				Capital measurement	
Syndicate aggregation:	VW [1]	Median [2]	Dominant [3]	Lead [4]	EW [5]	EW [6]
$Tier\ 1\ Capital/RWA_{t-1}$	-4.323*** (0.548)	-0.962*** (0.438)	-2.083*** (0.410)	-0.563*** (0.169)		
$Tier\ 1\ Gap_{t-1}$					-1.830*** (0.452)	
$Total\ Capital/RWA_{t-1}$						-1.349*** (0.304)
Bank controls	Y	Y	Y	Y	Y	Y
Loan controls	Y	Y	Y	Y	Y	Y
Year fixed effects	Y	Y	Y	Y	Y	Y
Observations	29,121	29,121	29,121	29,121	29,121	29,121
R^2	0.542	0.545	0.549	0.540	0.419	0.419

This table shows robustness checks for the effects of bank regulatory capital for loan acquisition by nonbanks. The unit of observation in each regression is a loan-year. The dependent variable is the fraction of the loan held by nonbanks. Columns [1] to [4] use alternative methods to aggregate independent variables—bank controls shown in Table 3—up to the loan syndicate level. In particular, we consider the loan share value-weighted average, the median value, the simple average among the three ("dominant") banks with the largest loan shares, and the lead arranger's bank characteristics. Columns [5] and [6] examine alternative measures of bank regulatory capital as independent variables. The sample period is from 1993 to 2014. Where indicated, the columns include controls for bank, loan, and year fixed effects, and loan controls (a regulatory pass/fail dummy and the natural logarithm of loan maturity). All variables are defined in Table A1. Standard errors (in parentheses) are clustered at the year level. ***, **, and * denote 1%, 5%, and 10% statistical significance, respectively.

5.7 Table 6: Nonbank-entry robustness

Read: Table 6 shows that the nonbank-entry result is robust to different ways of aggregating bank capital inside a loan syndicate: equal-weighted average, value-weighted average, median, dominant-bank average, and lead-arranger capital. The relation is strongest when focusing on banks with larger loan shares.

Economic message: The banks with the most exposure to a loan matter more for nonbank entry. This is intuitive: if the largest bank holders are capital constrained, their sales create more room for nonbank buyers.

5.8 Tables 7 and 8: Basel III shock

Table 7
Summary stats for Basel III capital shortfall tests

	<i>N</i>	Mean	Std.	p25	Med.	p75	<i>N</i>	Mean	Std.	p25	Med.	p75	Raw diff.	[Norm. diff.]
	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]	[10]	[11]	[12]	[13]	[14]
Loan-level variables														
<i>Loan Sale</i>	34,648	0.025	0.155	0	0	0								
<i>Loan Share/Assets</i>	34,648	0.125	0.148	0.028	0.075	0.160								
<i>Loan Size</i>	34,648	\$2.1	\$87.0	\$15.0	\$30.0	\$700.0								
<i>Agent Bank</i>	34,648	0.164	0.370	0	0	0								
Bank-level variables														
	Below-median capital shortfall						Above-median capital shortfall							
<i>Basel III Tier 1 Shortfall</i>	125	-0.043	0.009	-0.050	-0.040	-0.036	126	-0.020	0.007	-0.025	-0.021	-0.016	-0.023	[-2.76*]
<i>Tier 1 Capital/RWA</i>	125	0.149	0.031	0.125	0.145	0.172	126	0.131	0.028	0.111	0.129	0.146	0.018	[0.61*]
<i>Bank Size</i>	125	22.25	1.555	21.22	22.08	23.07	126	22.11	1.883	20.80	21.52	22.81	0.990	[0.03]
<i>Wholesale Funding</i>	125	0.187	0.091	0.130	0.174	0.217	126	0.184	0.092	0.123	0.161	0.228	0.003	[0.02]
<i>Real Estate Loan Share</i>	125	0.685	0.192	0.617	0.743	0.845	126	0.674	0.181	0.600	0.706	0.825	0.011	[0.03]
<i>CM Loan Share</i>	125	0.206	0.120	0.113	0.160	0.261	126	0.201	0.115	0.128	0.173	0.242	0.005	[0.04]
<i>Non-Interest Income</i>	125	0.264	0.169	0.160	0.235	0.318	126	0.246	0.150	0.153	0.220	0.290	0.018	[0.12]
<i>Return on Assets</i>	125	0.004	0.004	0.003	0.004	0.006	126	0.004	0.003	0.003	0.004	0.006	0.000	[0.07]
<i>Loan Loss Provisions</i>	125	0.002	0.003	0.000	0.001	0.002	126	0.001	0.001	0.000	0.001	0.002	0.001	[0.17]
<i>Allowance for Loan Losses</i>	125	0.000	0.000	0.000	0.000	0.000	126	0.000	0.000	0.000	0.000	0.000	0.000	[0.12]
<i>Average Loan PD</i>	125	0.045	0.110	0.001	0.010	0.031	126	0.070	0.195	0.002	0.008	0.035	0.030	[0.20]

The sample is restricted to loans held by at least two U.S. banks with valid covariates as of 2012:Q2. Loan-level variables are averaged (unweighted) across loan share-years. Bank level variables are averaged across bank years. Bank-level summary statistics split by above- and below-median *Basel III Tier 1 Shortfall* as of 2012:Q2. The sample includes data from 2012:Q2 and 2012:Q3. Differences in means (row) are reported in columns [13]. Normalized differences are reported in column [14]. We indicate normalized differences in excess of 0.25 with a “*” as per the taberns and Rubin (2007) rule of thumb. All variables are defined in Table A1.

Table 8
Basel III capital shortfall and bank loan sales

Dependent variable:	$\Delta \text{Basel III Tier 1 RWA}_i$		<i>Loan Sale_{itj}</i>				$\Delta \text{Nonbank Share}_i$	
			Exclude FIRE	No. amend	Exclude fixed effects	Alternative measurement		
	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]
<i>Basel III Tier 1 Shortfall</i>	-0.122*** (0.041)	-0.362*** (0.135)	-0.406*** (0.147)	-0.462*** (0.150)	-0.491*** (0.133)	-0.279*** (0.138)	-0.098** (0.044)	
<i>High MSR Exposure</i>						0.014*** (0.003)	0.006*** (0.002)	
<i>Tier 1 Capital/RWA</i>	-0.164*** (0.021)	-0.295*** (0.068)	-0.307*** (0.073)	-0.340*** (0.079)	-0.348*** (0.066)	-0.083 (0.051)	-0.143* (0.077)	0.008 (0.155)
Loan controls	N/A	N/A	N/A	N/A	N/A	N/A	N/A	Y
Loan fixed effects	Y	Y	Y	Y	Y	Y	Y	N/A
Observations	838	218,252	188,932	143,345	218,252	218,252	218,252	2,121
<i>R</i> ²	0.107	0.156	0.134	0.122	0.046	0.136	0.136	0.994

This table shows the effects of the 2012:Q2 proposed changes in bank capital regulation under Basel III for loan sales. The independent variable of interest, *Basel III Tier 1 Shortfall*, measures the bank level difference between the current (under Basel I) and proposed level of Tier 1 regulatory capital under Basel III. In column [1] the unit of observation in each regression is a bank. The dependent variable is the change in the Tier 1 capital ratio under Basel III from 2012:Q2 to 2013:Q2. In column [2] to [6], the unit of observation in each regression is a loan share-bank double. The dependent variable is an indicator variable equal to one if a lender reduces its ownership stake in a loan in 2012:Q3 that it funded in 2012:Q2. Column [3] excludes loans made to finance, insurance, and real estate. Column [4] restricts the sample to loans for which no contract amendment or refinancing took place during 2012:Q2 or 2012:Q3. Column [5] drops loan fixed effects from the estimation. Columns [6] and [7] use the above-median value of mortgage servicing rights (*High MSR Exposure*) as an alternative measure of banks' capital shortfall. Columns [8] and [9] use the loan syndicate level change in the nonbank share as the dependent variable and aggregate the independent variables in the syndicate level (i.e., maximum capital shortfall and MSR exposure dummies). Where indicated, columns control for loan fixed effects, and indicate variables for whether loans share are held by nonbanks or foreign banks. All columns include the bank control variables shown in Table 5, as well as *Loan Sale Propensity*, given by the fraction of loan shares sold per quarter, time-averaged between 2009:Q4 and 2012:Q2. Where indicated, columns include loan controls (loan-level average default probability and log loan maturity). All variables are defined in Table A1. Where “N/A” is shown, this indicates that the controls in question cannot be included. Standard errors (in parentheses) are clustered at the bank, loan, and year levels in columns [1], [7] to [7], and [8] to [9], respectively. ***, **, and * denote 1%, 5%, and 10% statistical significance, respectively.

Read: Table 7 describes the Basel III shortfall sample. Banks with larger shortfalls are hit harder by the U.S. implementation of Basel III. The table shows substantial heterogeneity in shortfalls, while syndicated-loan portfolio default probabilities are not systematically different across shortfall groups.

Read: Table 8 then uses the shock. Banks with larger Basel III Tier 1 shortfalls increase capital afterward, sell more loan shares, and are associated with larger increases in nonbank share at the syndicate level. A high mortgage-servicing-rights exposure indicator produces similar results.

Economic message: The Basel III evidence addresses the main omitted-variable concern. The results are not only correlations between low capital and loan sales. When a regulatory change unexpectedly worsens some banks’ capital positions, those banks respond by selling loans, and nonbanks fill the gap.

5.9 Table 9: Crisis sample

Read: Table 9 summarizes the sample used for the crisis tests. In the full SNC sample of loans outstanding in 2006:Q4, the average credit growth measure from 2006 to 2008 is strongly negative, and the loan exit rate is about 66.1%. The average precrisis nonbank share is 14.0%. In the smaller LSTA matched sample used for price tests, the average loan price decline is 8.8 percentage points.

Economic message: The crisis was a severe stress test. The question is whether loans with more nonbank funding before the crisis were more vulnerable during this period.

5.10 Table 10: Nonbank funding and credit availability

Read: Table 10 shows that higher precrisis nonbank share predicts lower credit growth and higher loan exit during the crisis. The effect is driven by unstable nonbanks. The stable nonbank share is not significantly associated with lower credit availability. At the firm level, the result remains, which means borrowers do not fully replace lost credit with other syndicated loans.

Economic message: The result supports a funding-fragility channel. Shadow-bank funding is not automatically fragile, but funding from unstable nonbanks is associated with credit contraction in bad times.

Table 9
Summary statistics for tests on real effects during the crisis

Panel A: Full SNC sample						
	<i>N</i>	Mean	Std.	p25	Med.	p75
	[1]	[2]	[3]	[4]	[5]	[6]
Loan characteristics						
<i>Credit Growth</i> ₂₀₀₈	6,439	-1.342	0.945	-2	-2	-0.091
<i>Exit</i> ₂₀₀₈	6,439	0.661	0.473	0	1	1
<i>Remaining Maturity</i> _{2006:Q4}	6,439	3.809	1.787	2	4	5
<i>Syndicate Size</i> _{2006:Q4}	6,439	16.67	35.87	4	7	14
<i>Loan Size</i> _{2006:Q4}	6,439	311.3	676.9	50	112	300
<i>Non-Pass</i>	6,439	0.058	0.234	0	0	0
Syndicate member characteristics						
<i>Nonbank Share</i> _{2006:Q4}	6,439	0.140	0.245	0	0	0.163
<i>Unstable Nonbank Share</i> _{2006:Q4}	6,439	0.020	0.062	0	0	0
<i>Stable Nonbank Share</i> _{2006:Q4}	6,439	0.006	0.037	0	0	0
<i>Tier 1 Capital/RWA</i> _{2006:Q4}	6,439	0.088	0.038	0.080	0.081	0.086
<i>Bank Size</i> _{2006:Q4}	6,439	19.63	1.601	18.30	20.74	20.80
<i>Wholesale Funding</i> _{2006:Q4}	6,439	0.420	0.074	0.380	0.430	0.433
<i>Real Estate Loan Share</i> _{2006:Q4}	6,439	0.505	0.135	0.424	0.569	0.569
<i>C&I Loan Share</i> _{2006:Q4}	6,439	0.125	0.055	0.091	0.099	0.140
<i>Non-Interest Income</i> _{2006:Q4}	6,439	0.230	0.189	0.128	0.146	0.484
Panel B: SNC-LSTA subsample						
	<i>N</i>	Mean	Std.	p25	Med.	p75
	[1]	[2]	[3]	[4]	[5]	[6]
Loan characteristics						
Δ <i>Loan Price</i> ₂₀₀₈	116	-0.088	0.072	-0.118	-0.070	-0.041
<i>Loan Price</i> _{2006:Q4}	116	0.979	0.024	0.973	0.986	0.992
<i>Remaining Maturity</i> _{2006:Q4}	116	3.664	1.157	3	4	4.5
<i>Syndicate Size</i> _{2006:Q4}	116	6.637	5.264	3	5	9
<i>Loan Size</i> _{2006:Q4}	116	761	1,130	148	346	861
<i>Non-Pass</i>	116	0.198	0.400	0	0	0
Syndicate member characteristics						
<i>Nonbank Share</i> _{2006:Q4}	116	0.453	0.344	0.119	0.398	0.837
<i>Unstable Nonbank Share</i> _{2006:Q4}	116	0.095	0.112	0	0.057	0.147
<i>Stable Nonbank Share</i> _{2006:Q4}	116	0.018	0.032	0	0	0.024
<i>Tier 1 Capital/RWA</i> _{2006:Q4}	116	0.105	0.051	0.079	0.083	0.102
<i>Bank Size</i> _{2006:Q4}	116	18.83	1.169	18.18	18.89	19.39
<i>Wholesale Funding</i> _{2006:Q4}	116	0.421	0.041	0.396	0.415	0.445
<i>Real Estate Loan Share</i> _{2006:Q4}	116	0.260	0.078	0.221	0.248	0.283
<i>C&I Loan Share</i> _{2006:Q4}	116	0.150	0.078	0.107	0.140	0.187
<i>Non-Interest Income</i> _{2006:Q4}	116	0.154	0.031	0.136	0.153	0.174

The unit of observation in each panel is a loan. Panel A describes data for the full SNC sample of loans outstanding as of 2006:Q4. Panel B describes data for the subsample of SNC loans matched with the LSTA data. Syndicate member characteristics are measured as of 2006:Q4 and equally weighted across all banks in the syndicate. Loan-level variables are measured as of 2006:Q4, except for *Non-Pass*, which is measured over 2007 and 2008. All variables are defined in Table A1.

Table 10
Nonbank loan share and credit availability during the crisis

Panel A: Credit growth in 2008							
Dependent variable: <i>Credit Growth</i> ₂₀₀₈							
	Loan-level					Firm-level	
	[1]	[2]	[3]	[4]	[5]	[6]	[7]
<i>Nonbank Share</i> _{2006:Q4}	-0.428*** (0.064)					-0.596*** (0.111)	
<i>Unstable Nonbank Share</i> _{2006:Q4}		-0.768*** (0.226)		-0.791*** (0.231)	-0.949*** (0.308)		-0.864** (0.377)
<i>Stable Nonbank Share</i> _{2006:Q4}			0.406 (0.574)	0.499 (0.562)	-0.340 (0.545)		0.189 (0.622)
Loan/firm controls	Y	Y	Y	Y	Y	Y	Y
Bank controls (synd. avg.)	N	N	N	N	Y	Y	Y
Observations	6,439	6,439	6,439	6,439	5,204	4,649	4,504
R ²	0.092	0.086	0.084	0.087	0.120	0.133	0.134
Panel B: Exit rate in 2008							
Dependent variable: <i>Exit</i> ₂₀₀₈							
	Loan-level					Firm-level	
	[1]	[2]	[3]	[4]	[5]	[6]	[7]
<i>Nonbank Share</i> _{2006:Q4}	0.225*** (0.050)					0.263*** (0.050)	
<i>Unstable Nonbank Share</i> _{2006:Q4}		0.408*** (0.164)		0.418*** (0.113)	0.553*** (0.147)		0.439*** (0.164)
<i>Stable Nonbank Share</i> _{2006:Q4}			-0.171 (0.295)	-0.220 (0.287)	0.141 (0.302)		-0.122 (0.297)
Loan/firm controls	Y	Y	Y	Y	Y	Y	Y
Bank controls (synd. avg.)	N	N	N	N	Y	Y	Y
Observations	6,439	6,439	6,439	6,439	5,204	4,649	4,504
R ²	0.034	0.086	0.084	0.086	0.118	0.133	0.133

This table examines the effects of nonbank loan funding for the credit availability during the crisis at the loan and firm levels. In columns [1] to [5] ([6] and [7]), the unit of observation in each regression is a loan (firm). In panel A, the dependent variable is the symmetric credit growth rate defined as the difference between credit in 2008:Q4 and 2006:Q4 divided by the average of credit in 2008:Q4 and 2006:Q4. Credit at the firm level sums across all term loans to a given firm. In panel B, the dependent variable is a loan exit dummy, which is equal to one if the loan (present in 2006:Q4) has exited the SNC sample by 2008:Q4. Exit at the firm level requires that, by the end of 2008, all of the firm's loans from 2006 have exited and the firm does not receive any new loans. Nonbanks with unstable liabilities include broker-dealers, hedge funds, and other investment funds, and nonbanks with stable liabilities include insurance companies and pension funds. Loan controls include loan size, syndicate size, borrower industry, the (log) remaining maturity, and an indicator variable for whether the loan is downgraded by the regulator in either 2007 or 2008. Loan control variables are measured as of 2006:Q4, except for *Non-Pass*, which is measured over 2007 and 2008. In columns [6] and [7], we instead include firm-level controls consisting of industry fixed effects and the total volume of credit as of 2006:Q4. Where indicated, columns include the bank controls shown in Table 3 (equal-weighted average across syndicate members and measured as of 2006:Q4). All variables are defined in Table A1. Heteroscedasticity-robust standard errors clustered at the industry-level are reported in parentheses. ***, **, and * denote 1%, 5%, and 10% statistical significance, respectively.

5.11 Figure 5: Nonbank liability structure and loan prices

Read: Figure 5 plots loan prices during 2007–2009 for loans with above- and below-median stable or unstable nonbank funding. The top panel shows little price separation by stable nonbank share. The bottom panel shows large price separation by unstable nonbank share, especially during 2008.

Economic message: The figure visually supports the idea that liability fragility matters. Loans held by unstable nonbanks experience larger price declines, consistent with selling pressure or reduced willingness to absorb illiquid credit risk during stress.

5.12 Table 11: Nonbank funding and price drops

Read: Table 11 estimates the price decline regressions. Nonbank share is negatively associated with the 2007–2008 loan price change. With controls, a one-standard-deviation increase in nonbank share is associated with a 1.69 percentage point larger price decline, which accounts for about 19.2% of the mean loan price fall. When nonbanks are split by liability type, the unstable nonbank share has a large negative coefficient; the stable nonbank share is not significant.

Economic message: The price result complements the credit-availability result. Unstable nonbank participation is associated not only with less credit but also with more secondary-market price volatility.

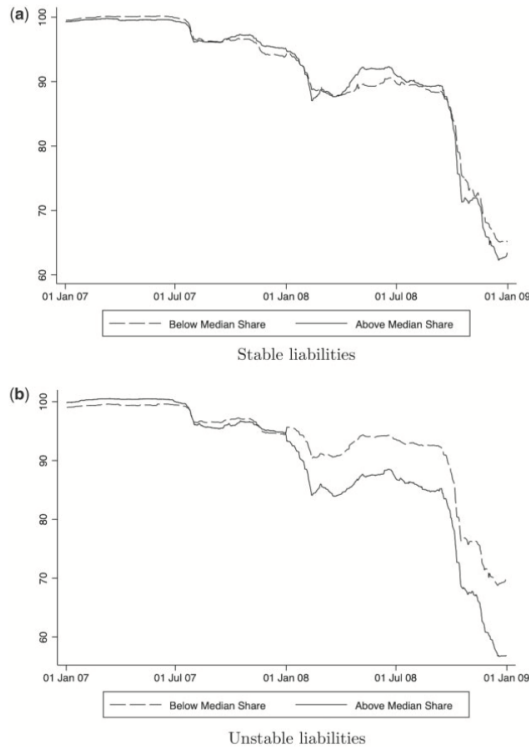


Figure 5
Nonbank liability structure and loan prices (2007–2009)
Average price (bid-ask midpoint) among traded syndicated term loans with above (solid) and below (dashed) median nonbank share in each category. The figure classifies syndicates according to whether nonbank syndicate members have stable (top panel) or unstable (bottom panel) liabilities. Nonbanks with stable liabilities are pension funds and insurance companies. Nonbanks with unstable liabilities are hedge funds, private equity, broker/dealers, and mutual funds. Nonbank share is the ratio of nonbank investment to total loan commitment.

Table 11
Nonbank loan share and price drop in the crisis
Dependent variable: $\Delta \text{Loan Price}_{2008}$

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Nonbank Share</i> _{2006:Q4}	-0.084*** (0.023)	-0.049** (0.019)						
<i>Unstable Nonbank Share</i> _{2006:Q4}			-0.236*** (0.061)	-0.182** (0.090)				
<i>Stable Nonbank Share</i> _{2006:Q4}					-0.237 (0.243)	0.006 (0.302)	-0.230*** (0.064)	-0.182** (0.091)
<i>Non-Fund</i>		0.012 (0.019)	0.003 (0.017)	-0.011 (0.020)	0.001 (0.018)	-0.012 (0.019)	0.005 (0.016)	-0.011 (0.018)
<i>Log(Remaining Maturity)</i> _{2006:Q4}		0.006 (0.005)	0.007* (0.005)	0.015** (0.006)	0.004 (0.005)	0.011* (0.006)	0.009** (0.005)	0.015** (0.006)
<i>Loan Price</i> _{2006:Q4}		1.468*** (0.329)	1.344*** (0.334)	1.466*** (0.382)	1.414*** (0.352)	1.600*** (0.365)	1.350*** (0.326)	1.466*** (0.387)
Bank controls (synd. avg.)	Y	Y	Y	Y	Y	Y	Y	Y
Observations	116	116	116	79	116	79	116	79
R ²	0.238	0.458	0.511	0.569	0.443	0.532	0.512	0.569

This table examines the effects of nonbank loan funding for the change in the secondary market loan price. The unit of observation in each regression is a loan. The dependent variable is the change in the loan price level between 2007 and 2008. The price level is measured as the average bid-ask midpoint. Nonbanks with unstable liabilities include broker-dealers, hedge funds, and other investment funds, and nonbanks with stable liabilities include insurance companies and pension funds. Columns (4), (6), and (8) require stable and unstable nonbanks to be present in the syndicate for sample inclusion. Bank-level variables are averaged across all bank syndicated members (equally weighted) as of 2006:Q4. Loan-level variables are measured as of 2006:Q4, except for *Loan Price*, which is measured over 2007 and 2008. Where indicated, columns include the bank controls shown in Table 3 (equal-weighted average across syndicate members). All variables are defined in Table A1. Heteroscedasticity-robust standard errors clustered at the industry-level are reported in parentheses. ***, **, and * denote 1%, 5%, and 10% statistical significance, respectively.

Table 12
Further evidence on term loan trading activity

Panel A: Role of bank capital
Sample:

	All trades			No amendments		
	Sellers	Buyers	Raw diff. [Norm. diff.]	Sellers	Buyers	Raw diff. [Norm. diff.]
	[1]	[2]	[3]	[4]	[5]	[6]
Crisis (2008)						
<i>Tier 1 Capital/RWA</i> _{2007:Q4}	0.087	0.097	-0.010+ [0.353]	0.087	0.098	-0.011+ [0.348]
<i>N</i>	1,069	1,179		541	361	
Pre-crisis (2007)						
<i>Tier 1 Capital/RWA</i> _{2006:Q4}	0.090	0.091	-0.001 [0.054]	0.091	0.091	0.000 [0.031]
<i>N</i>	701	1,186		300	308	

Panel B: Stable and unstable nonbank trading activity

Timing:	Crisis (2008)			Pre-crisis (2007)		
	Stable [1]	Unstable [2]	Diff. [3]	Stable [4]	Unstable [5]	Diff. [6]
Loans sold _t /holdings _{t-1} (%)	6.50	9.86	-3.36	6.73	6.87	-0.14
Loans bought _t /holdings _{t-1} (%)	13.18	9.20	3.98	6.16	7.93	-1.77
Number of sells	316	1,355		191	583	
Number of buys	641	1,265		175	673	

The table describes the identity buyers and sellers of term loan shares during the crisis (2008) and immediately prior to the crisis (2007). Panel A considers measures of bank Tier 1 capital for all buy and sell transactions by banks. A transaction is classified as a loan share sale (buy) whenever a bank that was (was not) in the syndicate in the previous year is not (is now) present this year. “No amendments” excludes transactions in years where the loan contract is amended. Each cell shows the average characteristic of the banks engaged in a loan share transaction as either sellers or buyers. A simple average is taken across loan transactions. The number of loan transactions (*N*) is indicated. The difference in the mean characteristic for each transaction type is indicated. Raw and normalized differences are reported in columns [3] and [6]. We indicate normalized differences in excess of 0.25 with a “+” as per the Imbens and Rubin (2007) rule of thumb. Panel B describes secondary market trading activity by nonbanks in the aggregate. As before, stable nonbanks include insurance companies and pension funds, and unstable nonbanks include broker-dealers, hedge funds, and other investment funds. Each cell shows the aggregate characteristic of the nonbank group engaged in a loan share transaction as either sellers or buyers.

5.13 Table 12: Who buys during the crunch?

Read: Table 12 studies buyer and seller identity during 2007 and 2008. In 2008, banks buying loan shares have higher Tier 1 capital than banks selling loan shares. Stable nonbanks buy more and sell less relative to unstable nonbanks during the crisis.

Economic message: Well-capitalized banks and stable nonbanks provide some liquidity during the crisis. But they do not fully offset the selling pressure from fragile nonbanks. This helps explain why nonbank-funded loans experience larger price drops.

6 Short summary

6.1 What the paper shows

- Bank capital regulation affects the composition of credit supply.
- Low-capital banks sell syndicated term loan shares in the secondary market.
- The effect is stronger when funding conditions are tight and when loans have worse regulatory ratings.
- Nonbanks absorb a meaningful part of the credit risk that banks shed.
- Surprise features of the U.S. Basel III implementation provide regulatory-shock evidence that supports the capital channel.
- During the 2008 crisis, loans with greater nonbank funding experience lower credit availability and larger price declines.
- These adverse crisis effects are concentrated among nonbanks with unstable liabilities.

6.2 Most important empirical designs

Design 1: loan-year fixed effects. Compare selling decisions across banks holding the same loan in the same year.

Design 2: Basel III shock. Use unexpected capital shortfalls from the U.S. Basel III implementation to identify bank capital pressure.

Design 3: crisis cross-section. Compare crisis outcomes for loans with different precrisis nonbank funding shares, especially stable versus unstable nonbank shares.

6.3 Most important economic result

The main result is not simply that shadow banks expanded. The main result is that a regulatory force affecting banks helps explain the expansion, and that the resulting risk migration can matter for financial stability.

6.4 Cleanest one-line takeaway

Bank capital regulation can push credit risk from banks to shadow banks; when the receiving shadow banks have fragile liabilities, this risk migration can reduce credit availability and amplify price volatility during crises.

7 Conclusion

Irani et al. provide evidence that capital regulation and shadow banking are tightly connected in the U.S. syndicated loan market. Banks with weaker capital positions sell more loans, especially when capital is scarce and when loans are more capital-intensive. Nonbanks absorb a substantial part of this credit risk. The paper's loan-year fixed effects design is powerful because it compares banks inside the same loan syndicate at the same time, and the Basel III implementation shock strengthens the causal interpretation.

The paper then shows why this migration of credit risk matters. Nonbank funding is associated with lower credit availability and larger loan price declines during the 2008 crisis, but only when the nonbanks have unstable liabilities. Stable nonbanks such as pension funds and insurance companies do not produce the same adverse patterns.

The policy implication is subtle. Higher bank capital requirements may make traditional banks safer, but they can also shift intermediation outside the regulatory perimeter. Whether this improves or worsens financial stability depends on who receives the risk and how fragile their funding is. The paper does not claim that nonbanks are always harmful. Instead, it shows that the distribution of credit across banks and shadow banks is a central part of the transmission of financial regulation.